Shohaib Ibne Monju

Email: shohaib.rafi25@gmail.com | LinkedIn | ResearchGate | Portfolio
Phone No: +8801795709898

Education

Bachelor of Science in Materials and Metallurgical Engineering *Feb 2020 – Mar 2025*
Bangladesh University of Engineering and Technology (BUET), Dhaka, Bangladesh
Relevant Courses: Electrical & Magnetic Properties of Materials, Phase Diagrams, Mechanical Behavior of Materials, Materials Processing for Micro & Nano-Systems, Computer Programming.
GPA: 3.18/4.00

Research Interests

- Machine Learning
- Density Functional Theory
- Energy Conversion
- Molecular Dynamics
- Energy Storage Devices
- Electrochemistry

Research Experience

Undergraduate Researcher (Remote) *Jan 2023 – Present*
PI & Group Lead: Dr. Saquib Ahmed, SUNY – Buffalo State University, New York, USA
Ahmed Research Group

Work Focus:
- Integrated machine-learning-based prediction models with materials datasets for structure–property analysis.
- Performed density functional theory (DFT) calculations to validate phase stability of high-entropy alloys.
- Conducted molecular dynamics (MD) simulations to study defect behavior and mechanical responses in multi-principal element alloys.

Tools & Methods: Python (pandas, scikit-learn, matplotlib), VASP (DFT), LAMMPS (MD), VESTA, OVITO, Atomsk, Packmol, MATLAB, OriginLab.

Current Research Project

Accelerated Design of High-Entropy Alloys via Probabilistic Substitution and Machine Learning with DFT Simulations
Collaborators: Abdul Hamid Rumman, Kaushik Barua, **Shohaib Ibne Monju**, Aabrar Bin Salahuddin, Md. Tohidul Islam, Saquib Ahmed
– Built ML classifiers (ANN, XGBoost) using elemental descriptors to predict HEA phase stability.
– Introduced a probabilistic substitution approach generating 50k+ HEAs; DFT confirmed stability of most.
Contribution: Writing – Original Draft, Methodology, Validation, Software, Formal Analysis.
Progress: Manuscript under review at *Journal of Alloys and Compounds.*

Undergraduate Thesis

Effect of Cryogenic Deformation on Properties of AA6063 Alloys
– Investigated DCT (-196°C) and rolling effects on AA6063; observed hardness up to 102.6 HV and UTS ≈ 392 MPa due to suppressed recovery and Orowan-type strengthening.
Supervisors: Dr. A. K. M. Bazlur Rashid

Capstone Project

Synthesizing Decaffeinating Filter from Cellulose Powder to Control Caffeine in Coffee *Dec. 2022 – Sep. 2023*
– Designed and synthesized a cellulose–tea composite filter using EGDMA crosslinker prepared from ethylene glycol and acrylic acid.

– Optimized pulp composition and pressing conditions to achieve caffeine removal efficiency.
Supervisor: Dr. Ahmed Sharif

Industrial Attachment

Industrial Training May 2024
Had the opportunity to visit DBL, a leading ceramics manufacturing company in Bangladesh.

- Observed raw materials and machinery for wall/floor tile manufacturing.
- Studied effects of raw materials on Modulus of Rupture (MOR).
- Identified common manufacturing flaws causing fractures and methods to mitigate them.

Publications

[1] Abdul Hamid Rumman, Kaushik Barua, **Shohaib Ibne Monju**, Mohd Rakibul Hasan Abed, Sadika Jannath Tan-Ema, Jafar F. Al-Sharab, Saquib Ahmed;
"Classification of CoCr-based magnetic thin films via GLCM texture features extracted from EFTEM images and machine learning," *AIP Advances*, vol. 14, no. 11, p. 115017, **Nov. 2024**. DOI

[2] Miah Abdullah Sahriar, Abdul Hamid Rumman, Ahammad Ullah, Kaushik Barua, **Shohaib Ibne Monju**, Munim Shahriar Jawad, Md. Atik Faisal, Ridwan Radit Ahsan, Houk Jang, Saquib Ahmed;
"Optical image analysis of WSe_2 thresholding for layer detection," *Computational Materials Science*, vol. 253, p. 113888, **May 2025**. DOI

Technical Skills

- **Experimental Skills:** Uniaxial Pressing, Sand Molding, Metallography, Microstructure Analysis, Chemical Analysis, Electrodeposition, Corrosion Testing.
- **Software & Computational Skills:** VASP (DFT), OriginLab, CES Edupack, MATLAB, SolidWorks, MS Office, LaTeX.
  Molecular Dynamics: Basic simulation via LAMMPS; structure generation / visualization with VESTA, OVITO, Atomsk, Packmol.
- **Programming Languages:** C, C++, Python (matplotlib, pandas, scikit-learn).

Language Proficiency

TOEFL: 110/120 (Reading: 30, Listening: 30, Speaking: 23, Writing: 27)

Standardized Test Scores

GRE: 311 (Quantitative: 164, Verbal: 147, AWA: 3.5)

Extra-Curricular Activities

- Vice President, BUET Career Club (2023–2024)
- Publication Secretary, SAMME (Students' Association of MME)
- Part-time Script Evaluator, Udvash Academic & Admission Care
- Instructor, ScholaraiD Academic & Admission Aid

References

Dr. Saquib Ahmed
Associate Professor, Engineering Technology
Founder and Director, Center for Integrated Studies in Nanoscience and Nanotechnology
The State University of New York, Buffalo State University
Email: ahmedsm@buffalostate.edu
faculty.buffalostate.edu/ahmedsm

Dr. Kazi Md. Shorowordi
Professor, Department of Materials and Metallurgical Engineering
Bangladesh University of Engineering and Technology (BUET), Dhaka, Bangladesh
Email: kazi@mme.buet.ac.bd
mme.buet.ac.bd/peoples/kazi-md-shorowordi/